PAGE 1: Compliance Management System — Mobile-First Field Redesign

A UX Research & Interaction Design Case Study

Tags: Product Discovery · UX Research · Interaction Design · Offline-First Design · Enterprise System

Role	**Platform**	**Domain**	**Year**
UX Designer & Product Analyst	iPad & iPhone (iOS)	B2B Enterprise Compliance System	2021

Introduction

This case study documents the end-to-end redesign of a legacy enterprise compliance management system for field-based users. The platform was mature and well-adopted on desktop, used by compliance managers, administrators, and office-based assessors. However, field users conducting on-site inspections, audits, and assessments at remote locations and live facilities had no way to use the system directly in the field. Without a portable, field-ready interface, assessors were forced into a two-step workaround: making notes outside the system on-site, then returning to a desktop to manually enter or update the assessment record. This broke the workflow at its most critical point — the moment of observation — and introduced delays, transcription errors, and compliance risk.

The strategic response was purpose-built field-native features that field assessors could carry into any environment and use to fill in assessment forms directly and offline, without needing to revert to pen and paper or defer data entry to a later desktop session. The offline-first architecture, draft save, and auto-sync features were not enhancements — they were the core enablers that made direct field use viable.

This case study covers the research, design, and validation of that redesign: diagnosing user pain, mapping journeys, designing for the field context, and validating through usability testing.

Approach: Research → Design → Validation

This project continues in two follow-on explorations: [Phase 2 — Product-Led Adoption] applies product-adoption analytics thinking to this redesign, and [Phase 3 — AI Enablement] specs out an AI copilot layer on top of it. Both are self-directed learning exercises built on this real project, not additional client work.

Prototype

(embed existing prototype video)

Skills Used

UX Research & Design: User Research (interviews, field observation) · Persona Development · Empathy Mapping · Customer Journey Mapping (As-Is & To-Be) · Workshop Facilitation · Wireframing · Interaction Design · Prototyping · Usability Testing (SUS scoring)

Product Thinking: Product Discovery · Feature Prioritisation · Roadmap Planning

Problem Statement

The Core Access Failure

Field assessors were conducting compliance inspections at remote sites, live facilities, and outdoor environments where a laptop or desktop was neither available nor practical. The compliance system had only a basic mobile interface, which produced a structural problem: the moment of assessment — when evidence was fresh, context was visible, and accuracy was highest — was exactly when the system could not be used, forcing assessors into a two-step workaround. The solution was to extend the system across multiple touchpoints — desktop, iPad, and mobile — with a seamless experience that allowed assessors to fill forms, work offline, and sync data without any break in workflow.

Two Costly Workarounds Emerged

Paper-Based Workaround	**Assessors took handwritten notes on-site and manually re-entered data at the desktop later. This introduced transcription errors, duplicated effort, and created compliance gaps between the moment of observation and the point of record.**
Deferred Completion	Assessors waited until returning to a fixed location to complete digital records, hours or

Paper-Based Workaround	**Assessors took handwritten notes on-site and manually re-entered data at the desktop later. This introduced transcription errors, duplicated effort, and created compliance gaps between the moment of observation and the point of record.**
	days after the site visit. Observations faded, context was lost, and time-sensitive incidents were flagged late.

The Quantified Usability Gap

Desktop Completion Rate	**82%, the baseline for a well-functioning system**
iPad Pre-Redesign Completion Rate	21%, 4× below desktop baseline
Primary Abandonment Point	Draft save and offline sync — the exact moments the field workflow required most

Problem Statement Summary

Field-based compliance assessors were unable to complete assessments at the point of inspection because the existing iOS interface was very basic and did not support offline capture, reliable draft save, or seamless sync — resulting in a 61-percentage-point gap in completion rate compared to desktop users, and creating compliance risk and data quality issues for field operations.

- Manual note-taking
- No offline access
- Frequent data loss
- Delayed validations
- Poor mobile compatibility

Research & Discovery Methodology

The research phase combined qualitative field investigation with quantitative funnel analysis. This dual approach was deliberate: qualitative methods revealed why assessors were failing; quantitative analysis revealed where and how severely. Together they built the complete picture needed to design a credible, evidence-driven solution.

Stakeholder Discovery Workshops	**Facilitated sessions with compliance managers, field assessors, and system administrators. Used feedback grids (Works Well / Needs to Change / Questions / Ideas) to surface competing priorities and hidden assumptions.**
Field Observation & User Interviews	On-site observation of assessors at work, watching how notes were taken, how the app was (or wasn't) used, and where confidence broke down. Interviews followed each observation to understand the emotional and cognitive experience.
User Personas & Empathy Mapping	Synthesised research into a primary persona (Adam, Field & Survey Officer) with empathy maps capturing what he does, thinks, feels, and needs at each stage of the assessment workflow.
Customer Journey Mapping (As-Is)	Mapped the current-state journey step-by-step, from travelling to a site through to post-submission, documenting actions, thoughts, feelings, and needs at each stage. Surfaced the emotional low points (anxiety, confusion, frustration) that corresponded to interface failures.
Usage Funnel Analysis	Analysed the 6-stage digital funnel from iPad access assignment through to full assessment submission, quantifying the gap between desktop and iPad completion rates and identifying the stage where drop-off was most severe (draft save / sync).
System Usability Scale (SUS)	Conducted SUS evaluations on both the desktop and the pre-redesign iPad interface. The desktop scored 76 (Good). The iPad pre-redesign scored 49 (Not Acceptable), a quantified confirmation of the usability gap.

Stakeholder Feedback Grid

✅ Works Well	⚠️ Needs to Change	❓ Questions	💡 New Ideas
On-field recording is easy and fast · Manual work eliminated · Templates auto-load based on location · Critical incident notifications visible · Frequent auto-saving	App performance is slow · Notifications are unintuitive · Syncing is confusing	How does communication work for non-iOS? · What are the offline data limits?	SMS/iMessage-based incident alerts · Gesture-based navigation · Voice-based search and incident recording · Dashboard with recent activities · Centralized notifications tab

(image: Feedback Grid)

Persona & Empathy Map — Adam (Field & Survey Officer)

(images: Persona, Empathy Map)

Stakeholder Assumptions and Risk Mapping

Assumption	Risk Level	Validation Approach
Users prefer mobile recording	High	Field interviews, analytics
SMS/iMessage is the best alert method	Medium	Survey, usage data
Offline sync is technically feasible	High	Dev partner consultation
Templates are reused across locations	Medium	Contextual inquiry
Users are tech-savvy and need minimal training	Low	Persona analysis

(image: Assumptions)

As-Is Customer Journey: The Failing Field Experience

Step	Does	Thinks	Feels	Needs
Travel to site, take notes	Uses pen & paper	"Hope I don't lose these notes."	Anxious, cautious	Mobile recording, auto-save
Choose template	Searches manually	"Which one fits this site?"	Confused, uncertain	Smart template suggestions
Fill out basic details	Enters manually	"This takes too long."	Frustrated	Autofill from location or history
Answer questionnaire	Scrolls through long forms	"I might miss something important."	Overwhelmed	Simplified UI, criticality alerts
Continue later	Saves and exits	"Will this sync correctly?"	Worried	Reliable offline sync, confirmation

(image: As-Is Scenario)

Key Friction Points from Field Research

- No reliable offline mode — one dropped connection lost all unsaved data
- Draft save was hidden and manual; most users didn't know it existed, auto-save was absent
- Sync was opaque — no visibility on whether data had synced; users re-entered data out of fear
- Desktop navigation ported to iPad — deep menus designed for mouse use were unusable with gloved or one-handed field operation
- Form inputs not optimised for touch — small tap targets, no keyboard auto-dismiss, date pickers unsuited to iOS
- No field-specific features — photo evidence, quick notes, and site context required separate apps

Design Solution: Offline-First, Field-Native

Design principle: the iPad interface must work reliably in the worst-case field scenario — no connectivity, one hand occupied, gloves possible, time pressure real. Everything else follows from that constraint.

From As-Is to To-Be: The Redesigned Journey

Stage	Does	Thinks	Feels	Gains
Travel to site	Opens app, auto-loads template	"This is fast and accurate."	Confident, focused	Location-based template suggestions
Fill out details	Autofill from GPS and history	"I'm saving so much time."	Relieved	Smart autofill, reduced manual entry
Record assessment	Uses offline mode with auto-save	"No worries about data loss."	Secure, efficient	Offline recording, auto-save
View alerts	Receives SMS/iMessage notifications	"I can act immediately."	Empowered	Real-time criticality alerts
Submit & sync	Syncs automatically when online	"Everything's backed up."	Satisfied	Seamless sync, confirmation feedback

(image: To-Be Scenario)

Before & After: Key Design Changes

Before	After
No offline mode, data lost on disconnection	Offline-first architecture: assessments work fully without any connectivity
Manual, hidden draft save	Auto-save every 60 seconds + persistent draft recovery on relaunch

Before	After
Opaque sync, no user feedback	Sync status bar: Saved Offline → Syncing… → Submitted, visible at all times
Desktop navigation (5+ taps to reach assessment)	Flat field navigation, assessment launch in 2 taps from home screen
Small touch targets, desktop form controls	iOS-native controls, large tap targets, optimised for gloved/one-hand use
No photo or note capture within the flow	In-context photo attachment and quick notes embedded within each assessment field
No onboarding for new iPad users	Contextual first-use tooltips on offline mode, draft save, and sync

Feature Prioritisation: Phased Delivery

A staged capability roadmap grounded in field research and user feedback, prioritizing progressive empowerment for assessors — from basic mobile access to advanced interaction modes.

Short-Term Enhancements

- Mobile login, assessment creation, and resume flow
- Offline mode with auto-sync and criticality alerts
- Autosave and sync status visibility

Long-Term Vision

- Dashboard with recent activities and centralized notifications
- Gesture-based navigation for faster input
- Voice-enabled search and incident recording
- SMS/iMessage alerts for critical events

(images: experience-based roadmap, prioritisation grid, stakeholder map, users' hopes and fears)

Validation: Did the Redesign Work?

Metric	Before	After	What it validates
SUS Score (iPad interface)	49 (Poor)	81 (Excellent)	The interface itself became genuinely usable
Assessment Completion Rate (iPad)	21%	55%	Assessors could now complete the task the redesign targeted
Time on Task (first submission)	~62 min	~28 min	The redesigned flow removed unnecessary steps

Design Process

Wireframes (Balsamiq) — Explored multiple user journeys, iterated 2–3 versions, focused on usability and task efficiency. *(image: wireframes)*

High-Fidelity UI — Dashboard, Assessment List (Card & List views), Quick Filters & Search, Record Assessment, Assessment Summary. *(images: high-fidelity screens, mobile UI screens)*

PAGE 2: Product-Led Adoption

Activation, Feature Adoption & Retention Analysis

Tags: Product-Led Adoption (Enterprise Adaptation) · Activation & Funnel Analysis · Feature Adoption Analysis · Retention Modelling

> **This is a self-directed learning exercise, not client work.** Everything below — the funnel numbers, feature-adoption curve, and retention percentages — is illustrative data I constructed to practice product-adoption analytics (activation funnels, time-to-value, retention modelling) on top of the real Phase 1 case. It was not collected from, delivered to, or requested by the client. Treat this as "here's how

I'd think about adoption if I had this data," not "here's what happened." See Phase 1 — the real, delivered project →

"Product-Led Growth" is usually associated with self-serve, bottom-up software (Slack, Notion, Figma) where the product itself drives acquisition. This project doesn't have that — licenses were assigned top-down by administrators, not self-served. What does transfer cleanly from PLG thinking is the discipline of activation, time-to-value, feature adoption, and retention as measurable product metrics. This page applies that discipline to an already-adopted enterprise tool — product-led *adoption and retention*, not product-led acquisition.

Introduction

Phase 1 solved access — assessors could finally use the system in the field. Phase 2 asks the next question: once access was possible, did people actually adopt it, and did that adoption hold over time? Every UX decision from Phase 1 is translated here into a measurable product metric.

The Aha Moment: Two Valid Completion Paths

Path 1: Single Session	**A field user completes their first full assessment on iPad and submits it, experiencing the system working at the point of inspection for the first time.**
Path 2: Draft → Offline → Sync → Submit	User starts assessment → saves to draft (auto or manual) → completes offline → draft auto-syncs on reconnect → user reviews and submits. Typical for complex, multi-location assessments. Pre-redesign: effectively broken. Post-redesign: a primary viable route.

The redesign's primary success condition was making Path 2 as reliable as Path 1, because for the most demanding field assessments, Path 2 is the only realistic option.

The 6-Stage Activation Funnel

The funnel begins at iPad access assignment — existing enterprise users being extended to a new platform, not new sign-ups.

Funnel Stage	**Desktop**	**iPad Pre**	**iPad Post (Proj.)**	**Interpretation**
iPad Access Assigned	500 (100%)	300 (100%)	300 (100%)	Baseline — existing users given iPad access
First iPad Login	480 (96%)	228 (76%)	261 (87%)	iPad drop hints at device unfamiliarity
Assessment Opened on iPad	450 (90%)	150 (50%)	225 (75%)	Half of iPad users couldn't navigate to an assessment
Field Data Entered (any)	430 (86%)	108 (36%)	198 (66%)	Most couldn't enter data effectively on iPad
Draft Saved or Submitted	415 (83%)	78 (26%)	177 (59%)	Draft save broken pre-redesign; most abandoned
Assessment Fully Submitted ★	**410 (82%)**	**63 (21%)**	**165 (55%)**	**Aha moment — 4× gap pre-redesign; projected to close significantly**

Completion Path Breakdown

Completion Path	**Desktop**	**iPad Pre-Redesign**	**iPad Post-Redesign (Proj.)**
Single Session Complete	82%	21%	55%

Completion Path	Desktop	iPad Pre-Redesign	iPad Post-Redesign (Proj.)
Draft → Sync → Submit	10%	5%	38%
Abandoned	8%	74%	7%

Insight: The post-redesign projection shows a deliberate shift — the draft→sync→submit path grows from 5% to 38% of completions. The feature that drives activation is offline mode, not UI polish.

Feature Adoption (Post-Launch, 8 Weeks)

Feature	Week 1	Week 4	Week 8	Role
Offline Mode	41%	68%	83%	Aha Moment Core
Auto-Sync on Reconnect	38%	72%	88%	Aha Moment Core
Draft Save & Resume	45%	74%	86%	Aha Moment Core
Photo Attachment	22%	44%	61%	Secondary
Quick Notes	18%	38%	54%	Secondary
Offline Map	12%	28%	41%	Secondary

Retention

Sustained assessment submission rate over 30–90 days — not logins or time-in-app, but actual compliance output.

Metric	Before	After
Time to First Submission	~62 minutes	~28 minutes
Draft→Sync→Submit Path Usage	5%	38% of completions

Metric	Before	After
90-Day Sustained Submission Rate	17%	~64%

UX Work as Adoption Interventions

UX Work	Adoption Effect
Identified desktop inaccessibility through field research	Activation-blocker analysis — root cause of near-zero field completion rate
Designed offline-first assessment capture with draft save	Removed the single biggest barrier to Path 2 completion — the core aha-moment enabler
Built auto-sync with visible status indicator	Reduced abandonment anxiety — the sync-state indicator directly addresses the fear-of-data-loss barrier that caused mid-assessment dropout
Simplified iPad UI with field-only feature set	Reduced time-to-value — fewer navigation steps to reach submission
SUS scoring before and after redesign	Quantified UX improvement as a proxy for activation probability
Designed contextual first-use onboarding tooltips	Accelerated feature discovery — ensured offline mode and draft save were adopted early

Adoption Metric Framework

Concept	Applied Here
Activation (Aha Moment)	First full assessment completed and submitted on iPad
Time-to-Value	Pre-redesign: ~62 min with high abandonment. Post-redesign target: ~28 min
Retention	Sustained assessment submission rate over 30–90 days

Concept	Applied Here
Feature Adoption	Uptake of offline mode, draft save, and auto-sync — the three features that make field use viable

Recommendations

Instrument offline workflows specifically. Standard analytics cannot capture offline behaviour at the time of occurrence. A local event queue that timestamps and stores events offline, syncing on reconnect, gives the product team full visibility into the offline assessment journey — including where offline users drop off.

Surface draft recovery prominently on relaunch. The most anxiety-inducing moment for field users is returning to the app after a connectivity loss. The relaunch screen should proactively surface any pending drafts: "You have 1 assessment saved offline, ready to sync." This single intervention can meaningfully reduce abandonment at the sync stage.

Use completion rate as the core product health metric. Completion rate — not logins, not time-in-app — reflects actual compliance work completed. It's the most honest signal of whether the redesign is delivering real value to field users.

Continue to Phase 3: See how this adoption framework extends into an AI product specification — Compliance Field Copilot →

PAGE 3: AI Enablement — Compliance Field Copilot

AI Product Specification

Tags: Generative AI · Agentic Workflow Design · Product Analytics · Voice UX · B2B Enterprise SaaS · Offline-First

> **This is a self-directed learning exercise, not client work.** This page is a product specification and AI solution design I wrote to practice AI-enablement product thinking — agent architecture, tool design, guardrails, voice UX — on top of the real Phase 1 case. No part of this was requested by, built for, or delivered to NoMoBo's client. All metrics referenced here are illustrative, carried forward from Phase 2's

fictitious adoption data, not real production numbers. See Phase 1 — the real, delivered project → · See Phase 2 — the adoption-analytics exercise →

Role	Platform	Relationship	Status
Product Analyst & AI Solution Designer	iPad / iPhone / Web Admin	Extension of Phase 1 (real) and Phase 2 (illustrative)	Personal learning project — product specification only, not built

Separate product, shared foundation. Phase 1 solved the *access and workflow* problem — real, delivered, offline-first mobile redesign. Phase 2 explored an adoption-analytics lens on that redesign. Phase 3 asks: if I were designing an AI layer on top of this product, what would it look like? Compliance Field Copilot is that exploration — a distinct AI product layer that accelerates completion and reduces abandonment for the individual field assessor.

1. Executive Summary

Phase 1 established that field assessors failed to activate because the product could not be used at the point of inspection — producing a 61-percentage-point completion gap vs desktop users and 74% assessment abandonment on mobile. The redesign targeted offline mode, draft save, and auto-sync as the core activation enablers.

Phase 3 asks: once field access is viable, what intelligent assistance further compresses time-to-value and helps assessors complete work with confidence? The answer explored here is an embedded Compliance Field Copilot — an AI assistant that helps assessors select templates, capture evidence by voice, complete complex questionnaires, and submit with confidence, while giving compliance managers plain-language visibility into field adoption trends.

2. Problem Statement — What Phase 1 Left Unsolved

Residual friction	Even with offline sync, assessors still face template confusion, long forms, manual re-entry from handwritten notes, and anxiety about submission completeness.
Adoption blind spots	Standard analytics struggle to capture offline drop-off in real time — an assessor abandoning mid-assessment offline is invisible until they reconnect, by which point the moment (and often the data) is lost.

Residual friction	**Even with offline sync, assessors still face template confusion, long forms, manual re-entry from handwritten notes, and anxiety about submission completeness.**
Manager visibility	Compliance managers lack plain-language access to adoption trends and site-level risk across their teams.

3. Product Vision — Compliance Field Copilot

An in-context AI copilot that meets field assessors where work actually happens — on-site, offline, under time pressure.

Product name: Compliance Field Copilot (working title)

Positioning: Add-on AI module for the enterprise compliance platform. Deployed as an integrated layer within the Phase 1 mobile experience, with an admin-facing insights console for compliance managers.

Primary personas

Persona	AI need	Success signal
Field Assessor	Fast, accurate capture at inspection site; confidence data will sync	Higher completion rate; lower abandonment
Compliance Manager	Visibility into field adoption, risk gaps, and team performance	Faster issue detection; clearer team-level picture

4. AI Capabilities — Detailed Specification

4.1 Smart Template Recommender — Uses GPS location, facility type, assessment history, and the org's template library to rank the top 1–3 templates with confidence scores. *Example: "Based on your location at Warehouse B and your last 3 visits, I recommend the Fire Safety — Industrial template (92% match). Use this one?"*

4.2 Context-Aware Autofill — Pre-populates fields from prior assessments at the same site, org defaults, and device context. Assessor confirms before save — never silent overwrite.

4.3 Voice-to-Form Capture — Assessor speaks observations; speech-to-text + entity extraction maps content to structured form fields. *Example: "Fire exit blocked at loading bay B, severity high" → copilot proposes field values; assessor taps Confirm.* Directly attacks the transcription-error and double-entry abandonment drivers from Phase 1.

4.4 Guided Completion & Criticality Alerts — Surfaces next required critical items, explains why they matter, and flags incomplete sections before submit.

4.5 Pre-Submit Validation Agent — Runs validation rules before submit: required fields, evidence attachments, consistency checks vs prior visits. Returns a pass/fail with a fix list. Copilot recommends fixes; assessor retains submit authority.

4.6 Adoption Risk Intelligence (Admin Console) — ML models trained on Phase 2's illustrative funnel data predict activation risk (user unlikely to complete their first assessment within 14 days) and abandonment risk (mid-assessment dropout probability during offline/sync stages). Outputs: manager dashboard alerts and in-app nudges like "You have 1 draft ready to sync."

4.7 Manager Insights Copilot (Natural-Language Analytics) — LLM-RAG over assessment data lets managers ask plain-language questions and get concise answers with drill-down links. *Example queries: "Which sites had the lowest completion rate this month?" · "Where are assessors abandoning during offline sync?"*

5. Agent Architecture

A lightweight orchestrator with specialised tools, not a single monolithic chatbot.

Compliance Field Copilot (Orchestrator)

Intent routing · Session state · Guardrails · Disclosure

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Template Autofill Voice-to- Validation Adoption Manager

Recommender Engine Form NLP Agent Risk RAG Q&A

Scorer

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Platform APIs: assessments · drafts · sync · templates

Offline queue · audit log · manager escalation queue

Tool catalogue

Tool	**Input**	**Output**	**Autonomy**
get_site_context	GPS, account ID	Site profile, history	Automatic
suggest_templates	Site context	Ranked template list	Suggest → user confirms
autofill_fields	Template + history	Pre-filled draft	Suggest → user confirms
transcribe_voice_note	Audio blob	Structured field proposals	Suggest → user confirms
validate_submission	Draft assessment	Pass/fail + fix list	Advisory
score_adoption_risk	User behaviour events	Activation/abandonment risk score + nudge	Automatic (admin)
query_compliance_analytics	NL manager question	Answer + citations	Read-only

6. Sample Conversation Flows

Template selection (happy path) *Copilot:* Good morning. You're at Warehouse B. I recommend the Fire Safety — Industrial template based on this site and your recent visits. Start with this template? *Assessor:* Yes. *Copilot:* I've pre-filled site address, assessor ID, and last inspection date. Please confirm or edit. Ready to begin Section 1?

Voice capture → structured fields *Assessor (voice):* Loading bay fire exit partially blocked by pallets, high severity, photo attached. *Copilot:* I captured: Location — Loading Bay · Issue — Blocked fire exit · Severity — High · Evidence — 1 photo. Add to Section 4? *Assessor:* Confirm.

Offline sync reassurance *Copilot (on relaunch):* Welcome back. You have 1 assessment saved offline from yesterday — 85% complete. Sync now and finish the remaining 2 critical items?

Escalation — high-severity finding *Copilot:* This finding exceeds your organisation's auto-close threshold. I've flagged it for manager review and logged the draft. You can still submit — your manager will be notified automatically.

7. Adoption Metrics — AI Success Criteria (Illustrative)

Adoption Metric (Phase 2, illustrative)	**AI Intervention**	**Target Impact**
Activation rate (first completed assessment)	Template recommender + autofill + onboarding copilot	+15–20% field activation
Time-to-value	Autofill + guided completion	Additional 20–30% reduction beyond Phase 1
Draft→sync→submit path share	Offline relaunch prompts + validation agent	Maintain 38%+ path share; reduce sync-stage abandonment
Abandonment rate	Voice-to-form + pre-submit validation	Hold at ≤7% (vs 74% pre-redesign)
Retention (30–90 day submission rate)	Adoption-risk nudges	+10% sustained submission rate

North Star: Assessment completion rate (not logins). **Core:** Offline path reliability + AI-assisted capture. **Depth:** Predictive nudges that catch an assessor before they abandon mid-task.

8. Guardrails & Enterprise Governance

- **Human in the loop** — AI suggests; assessors confirm all field entries and submissions.
- **No auto-approval** — Copilot cannot certify compliance outcomes or override policy.
- **Audit trail** — Every suggestion, transcription, and validation logged with timestamp and user ID.
- **AI disclosure** — Copilot identifies as AI at session start.
- **Offline integrity** — Local queue preserves events; sync conflicts surfaced for manual resolution.
- **Manager escalation** — High-severity or policy-threshold breaches route to manager queue automatically.
- **Data minimisation** — Voice and location data retained per organisation policy; no training on client data without consent.

9. Technical Approach (High Level)

Layer	Approach
Field copilot UI	Embedded in Phase 1 iOS/iPad app; chat + voice FAB; offline-capable
Orchestration	Agent router with tool calling; session state per assessment draft
NLU / LLM	Intent classification + constrained generation for form-safe outputs
Voice	On-device or cloud STT; entity extraction to schema-bound fields
Adoption ML	Propensity models on event stream (activation, abandonment)
Manager RAG	Embeddings over metrics + assessment summaries; NL query interface
Integration	REST APIs to assessment, template, draft, sync, and notification services

10. Phased Rollout Roadmap (Hypothetical)

Phase	Scope	Adoption focus
3A — Foundation	Template recommender, autofill, guided completion	Activation + time-to-value
3B — Capture	Voice-to-form, pre-submit validation	Abandonment reduction
3C — Intelligence	Adoption-risk models, manager alerts, nudges	Retention (predictive engagement nudges)
3D — Admin	Manager Insights Copilot (NL analytics)	Team-level adoption visibility

11. Expected Impact (Illustrative)

- Further compression of field assessment time beyond Phase 1 UX gains

- Reduced compliance gap risk from missed critical items and transcription errors
- Higher sustained adoption per assessor, driven by proactive abandonment-risk nudges

12. Relationship Across Phases

Phase	**What it is**
Phase 1 (real)	UX research and offline-first redesign — making field use possible. Delivered for NoMoBo's client, 2021.
Phase 2 (illustrative)	Product-adoption analytics applied to Phase 1's outcome — a learning exercise in activation, funnel, and retention thinking.
Phase 3 (illustrative, this page)	AI copilot specification — a learning exercise in agentic product design, voice UX, and guardrails, built on Phase 2's fictitious adoption data.

Skills demonstrated: AI Product Design · Agentic Workflow Design · Conversational AI · Voice UX · Product Analytics Thinking · B2B Enterprise SaaS · Requirements Specification · Guardrails & Governance · Cross-functional Product Thinking